Algebraic Geometry WS 26/27

RECAP Sheet 4: 4.11-10.11

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R — Recall

1. Let X be a topological space. How is $\mathrm{Open}(X)$ a category ? What is a presheaf on X ?
2. Given a topological space X, what is the presheaf of continuous functions with values in $\mathbb{R}$? Is it a sheaf ?
3. What does the sheaf condition say for disjoint unions of opens ? What about increasing unions ?

E — Explore

1. Give your own examples of sheaves, and of presheaves that are not sheaves.
2. Let X be a discrete space. What is a sheaf on X ? Give examples of commutative rings A with discrete $\mathrm{Spec}(A)$. What if $X = \{0, 1\}$ with opens $\emptyset, \{1\}, X$? Also give examples of commutative rings A for which $\mathrm{Spec}(A)$ this space X (the Sierpinski space).
3. What is an example of a basis of topology that is not closed under finite intersections ?
4. What is an example of a sheaf $\mathcal{F}$ such that the restrictions $\mathcal{F}(X) \to \mathcal{F}(U)$ are not surjective ? Injective ?
5. What are some examples of non-principal opens in $\mathrm{Spec}(A)$?

C — Connect

1. How do sheaves in $\mathrm{CRing}, \mathrm{Ab}, \mathrm{Set}$ relate to one another ? More generally, how do the statements "the presheaf $\mathcal{F}$ is a sheaf" and "The presheaf $\hom(c, \mathcal{F}(-))$ is a sheaf" relate to one another ?
2. When are two opens $D(f), D(g)$ in $\mathrm{Spec}(A)$ disjoint ? When is one contained in the other ?
3. For an algebraically closed field k and an affine algebraic set V, what is the relation between V and $\mathrm{Spec}(\mathcal{O}(V))$? For a closed point x in the latter, what is $k(x)$?
4. Let A be a commutative ring and $f \in A$. How do $D(f)$ and $A[\frac{1}{f}]$ precisely relate to one another ? How about $V(I)$ and A/I, for an ideal $I \subset A$?

A — Apply

1. (a) Prove that the Sierpinski space $S = \{0, 1\}$ with $\{1\}$ as the only nontrivial open is sober.

   (b) Deduce that if $f : X \to S(X)$ is a sobrification, then f induces an isomorphism of

A — Apply (continued)

posets $f^{-1} : \mathrm{Open}(S(X)) \to \mathrm{Open}(X)$.

(c) Deduce that for V an affine algebraic set over an algebraically closed field k, sheaves on V are equivalent to sheaves on $\mathrm{Spec}(\mathcal{O}(V))$.

2. Let $\mathcal{F}$ be a presheaf on a topological space X, and $f : X \to Y$ a continous map. Describe the structure maps of a presheaf $f_*\mathcal{F}$ on Y defined by $f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}(U))$, and prove that if $\mathcal{F}$ was a sheaf, then so is $f_*\mathcal{F}$.

3. Suppose A is a commutative ring such that $\mathrm{Spec}(A)$ is discrete. Prove that $\mathrm{Spec}(A)$ is finite, and deduce that $A \cong \prod_{i=1}^{n} A_i$ for some commutative rings A_i whose Spec is a point.

4. Let A be a commutative ring. Characterize the opens in $\mathrm{Spec}(A)$ that are quasi-compact.

P — Present

Possible starting questions for the tutorial:

- Why, intuitively does $A[\frac{1}{f}]$ correspond to an open in $\mathrm{Spec}(A)$ and A/f to a closed subset ?
- Can any topological space arise as $\mathrm{Spec}(A)$ for some commutative ring A ?

The tutor may also ask about other central definitions, examples, arguments, or connections from this sheet.